

Class 19 Divisibility

Goals

1. Use the division algorithm repeatedly to find $\gcd(a,b)$
2. Construct proofs involving divisibility of integers.

Recall the Division Algorithm

For all a, n in the integers with $n > 0$
there exist unique integers q and r with
 $0 \leq r < n$ such that $a = nq + r$.

Definition: If a and n are integers with $n \neq 0$, we say n divides a if $a = nq$ for some $q \in \mathbb{Z}$.

other ways to say the same thing:

n is a factor of a . n is a divisor of a .
 a is a multiple of n a is divisible by n

Notation $n|a$ read n divides a

Examples $3|18$ since $18 = 3 \cdot 6$
 $5|20$ since $20 = 5 \cdot 4$

Definition: If a and b are nonzero integers, the greatest common divisor of a and b is the largest integer, k , that divides both a and b .

Notation $\gcd(a, b) = k$

Example $\gcd(12, 18) = 6$

$$\gcd(13, 18) = 1$$

Definition: Integers a and b are relatively prime (or coprime) iff

$$\gcd(a, b) = 1$$

Proof Problems

#8. The divides relation is linear.

If $n \neq 0$ and $n|a$ and $n|b$, then $n|ax+by$ for any integers x & y .

Proof:

Let $n|a$ and $n|b$. This means $a = nj$ and $b = nk$ for some integers j and k .

So for any integers x and y we have

$$ax + by = njx + nky = n(jx + ky).$$

Thus $n|ax+by$ since $jx+ky$ is in $\mathbb{Z}$ by closure of $\mathbb{Z}$ under addition and multiplication

Euclidean Algorithm to find gcd (a, b)

Find the gcd (704, 407)

$$704 = 1 \cdot 407 + 297$$

$$407 = 1 \cdot 297 + 110$$

$$297 = 2 \cdot 110 + 77$$

$$110 = 1 \cdot 77 + 33$$

$$77 = 2 \cdot 33 + \boxed{11}$$

$$33 = 3 \cdot 11 + 0$$

Euclidean

Algorithm

to find

gcd (a, b).

Repeated use
of the Division
Algorithm

Reversing this process, it is always possible to find integers x and y satisfying the equation $\text{gcd}(a,b) = ax + by$.

Example

Write 11 as a linear combination of 704 and 407.

$$11 = \underline{77} - 2 \cdot \underline{33}$$

$$= 77 - 2(110 - 77) = 3 \cdot \underline{77} - 2 \cdot \underline{110}$$

$$= 3(297 - 2 \cdot 110) - 2 \cdot 110 = 3 \cdot \underline{297} - 8 \cdot \underline{110}$$

$$= 3 \cdot 297 - 8(407 - 297) = 11 \cdot \underline{297} - 8 \cdot 407$$

$$= 11 \cdot (704 - 407) - 8 \cdot 407$$

$$= 11 \cdot \underline{704} - 19 \cdot \underline{407}$$